



MONKEY BUSINESS

News of The Funky Monkeys, Lynbrook High School Robotics, FIRST® Team 846



Table of Contents

- | | |
|----------------------------------|--|
| 1. Making an Impact | 4. First Culture |
| 2. Beyond the Bot | 5. Where Can You Find the Class of 2024? |
| 3. Freshman Machining Experience | |

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[Monkey Box](#)
[YouTube Videos](#)



Making an Impact



Khadija (jr.) speaks with the San Jose Girls Who Code Chapter.

Suzanne Das (soph.)

The Impact Award, the most prestigious honor in FIRST and a ticket to the World Championships, represents the culmination of years of our team's efforts in sharing our passion for robotics within the community. It's awarded to teams that best embody FIRST's mission to give young people the experience and skills to build a better world. A team submission consists of two parts; the first, an essay describing every initiative our team has taken to cultivate a connection with the community, and the second, an in-person presentation that highlights our coolest and most successful programs.

Each year, a trio of students pre-

pare a script and poster to present to the panel of judges. This year, I was one of the three presenters. Being an Impact Award presenter was a huge responsibility, one that I felt the full brunt of in the weeks leading up to our first regional. Thankfully, my role in the presentation was fairly simple: talk about the initiatives I was familiar with while the more experienced upperclassmen handle the rest. Much of what I spoke about in the presentation were initiatives that I've personally worked on, such as the annual demos we hold for girl groups, including the local Girls Who Code chapter and girls from Townley Grammar School in England.

Just hours before our first Impact

see Making an Impact, page 3

Beyond the Bot

How Non-Technical has Shaped my Robotics Experience.

Lucky Jain (sr.)

When I say robotics changed my life, it's not because I figured out I want to be an engineer (although that's what I would have said a month ago).

Scene: Tuesday, 7pm in room 608. An agenda is projected, and someone is talking from the front of the room. Twenty-some others are clustered in the back four rows. A few of them on their phones.

It's funny though because I love team meetings. My first one, at the end of sophomore year, was my first treasurer update. Yuvraj, the president at the time, asked me what our current balance was. I didn't know... embarrassment covered with nervous laughter and a twinge of excitement. Since then, I've taken notes: go up to the front of the room when you talk, volunteer for things, don't be late. And this, from my notes app two years ago:

Sell your idea, be enthusiastic.

It's funny when I tell people I'm in robotics, and they ask, "So what part of the robot did you work on?" *haha* I don't

see Beyond the Bot, page 4

The Machining Experience

Chenxia Liu (fr.)

After being introduced to Lynbrook Robotics by my friends at the start of the year, I was skeptical at first. Not having done any form of robotics before, I didn't know how I would contribute by joining the team and thought I would be more of a detriment than anything. However, I quickly found myself enjoying the welcoming atmosphere and the supportive people, and this team has become a significant part of my life.

At the beginning of the year, we were introduced to our teams' fall

I quickly found myself enjoying...the supportive people, and this team has become a significant part of my life.

workshops. Catered to new members, these workshops taught a variety of subjects, such as design and software. As I had just joined the team recently, I was very curious about the different parts of the



Chen (fr.) using the OMIO CNC router to cut out a part for the Pivot.

team and decided to attend most of the workshops the team had offered, such as the design workshops. I had very little experience with CAD (Computer Aided Design) software at this time, but the design workshop gave me the direction I needed to learn further. Additionally, I also attended electrical workshops and media workshops, where I could learn the

basics of crimping and soldering, as well as how to use Figma, a graphic design tool. This meant that while I didn't know very much about robotics before the fall workshops, I could still learn important technical skills and be a useful part of the team.

At the start of January, I had already become very familiar with the team and enjoyed being in it, and engaging in build season had only made me even more interested. A trove of opportunities had become available, such as helping with creating our scouting app or field elements, and one especially interesting one I was given was the chance to learn and create CAM (Computer Aided Manufacturing) programs. This gave me the ability to learn and use our CNC machines and bring plates created in CAD into real life and onto our robot. This was a skill I never thought I would ever have or was even aware of prior to joining robotics, and it's incredible that I could get the opportunity to do something like this, see *The Machining Experience*, page 4

FIRST Culture

An Interview with Anil Duvvuri.

Annav Kamatala (soph.)



The driveteams of Team 841, 846, and 8033 huddle together awaiting the results of close elimination match.

Coopertition® and Gracious Professionalism® means helping out and working with others, even if you're competing against them. It's about learning, teaching, and becoming better through working with peers. These Core Values are the cornerstone of the FIRST community showing how to preserve a culture of equity, diversity, and inclusion. I interviewed one of our senior members, Anil Duvvuri, to gain his perspective on what makes FIRST so special.

Q: Team shirt trading is like a massive and unique part of first culture see *FIRST Culture*, page 3

Making an Impact, Continued...

Award Presentation at the San Francisco Regional, the other presenters and I started preparing for the five-minute Q&A session that would follow our presentation. This was *Much of what I spoke about in the presentation were initiatives that I've personally worked on.*

the part I was most dreading. Our presentation was under our own control but questions were unpredictable and could be about anything, from the topics covered in the presentation to initiatives mentioned in our Impact essay. Answering questions required presenters to be familiar with all aspects of our

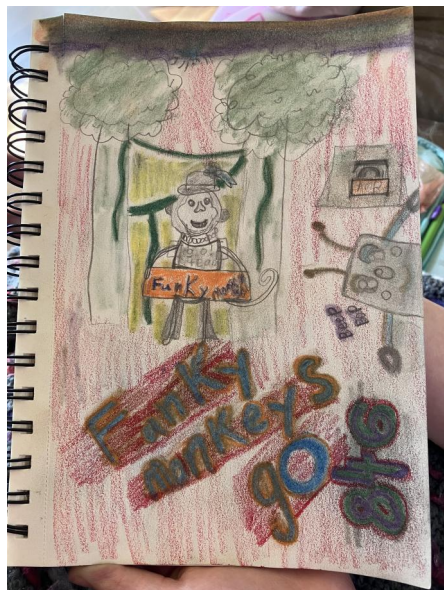


A student from our team-ran local middle school robotics club building a robot using a LEGO® SPIKE™ Prime kit.

team's outreach, and I couldn't confidently say I was. While carefully reading the essay, I not only gained a more holistic picture of our team's impact on the community but I also came across a few intriguing initiatives that have fallen

through over the years. One of these is our work introducing the FIRST LEGO League (FLL) to parents and middle schoolers across our district. For the two years I've been involved with our team's outreach, I had never once considered doing something along these lines but it interested me enough to dig up our old FLL presentation and try bringing this initiative back to life! Initiatives like this one jumpstart kids' journey into the world of STEM and robotics. Even if their interests steer them away from the STEM field, the collaborative and leadership skills gained from the experience are everlasting and applicable in all professions! 🤖

FIRST Culture, Continued...



ture. So what has been your favorite memory involving trading shirts?

A: Definitely, my favorite memory was trading a shirt with my friend from Team 841, the BioMechs, this year. I met him originally at SVR last year and now that we were in contact we decided to do a shirt trade. It was really great when we were able to join their Alliance in SFR and he was able to wear both our shirts at the same time. This

moment fostered a sense of unity and brought our teams closer together.

Q: Can you share some of your experiences interacting with other teams?

A: Interacting with teams flow skills, obviously at competitions you're going to be pet scouting and you're going to be with other teams generally. I was the technician for the Driveteam this year, you'd be talking with other teams drive teams as well and just interacting with them. You can kind of understand their thought processes what goes on in the robot but also what goes on like behind the scenes for them during their like outreach initiatives and stuff like that. So really, like learning from other teams and talking to them. This is a good way to like simply borrow as well.

Q: Why do you believe that fostering a friendly and welcoming community?

A: I would say it's just a good thing in general, but especially for FIRST, it helps learn a lot like if you're able to talk with people if you're good

standings, people make friends, you can learn a lot you can improve your team you can improve yourself even solid status having those connections. It's just beneficial for everyone.

Q: What advice would you give someone who's considering joining in FRC team but is unsure about the environment or what to expect?

A: I'd say it's a super supportive environment. Everyone's looking out for you. Everyone's everyone was there at one point and was a rookie and all the upperclassmen. They were rookies too, so they all know how to feel us so they're always going to be inclusive, and they're ready to help out.

The sense of unity, the joy of shared achievements, and the lessons we learn from each other are what make FIRST truly unique. As we continue to embrace these Core Values, we not only build better robots, but also a better community, where everyone has the opportunity to thrive and make meaningful connections. 🤖

Beyond the Bot, Continued...

have anything to say; I didn't touch the robot. And somehow robotics is still always on my mind, it's still my life, still my favorite.

Last week, we taught Arduino to little kids who've never seen circuits before. One student told us, "I want to be an engineer when I grow up." That event, that moment, was the culmination of everything I've worked on since I joined the non-technical side of the team.

It started at a SWOT meeting (it still amazes me how highly technical our "non-technical" side of the team is). Ms. Amin said something about how we should take the Girls Subsystem Challenge further and help girls outside our team. I spent that summer writing proposals, trying to get people on board, trying to figure out how to inspire people. Why did you join the team? Why did you stay? Why do you want to be an engineer? The people, the challenge, the scream-at-the-top-of-your-lungs feeling of earned success. Nontechnical showed me what our team is made of, and to use that to achieve anything I want.

Currently, I'm trying to require

everyone to join nontechnical, but it's not because we need more manpower (which we do). Mr. G said something recently: "This is not a company. This is education." We can't grow leaders in STEM if all they can do is build a robot. Hey, I loved learning about involute gears, four-bar linkages, and over-centering. But I'm who I am today because of nontechnical, what I learned about people, leading them, about myself, about perseverance, about caring.

More than the 3am nights trying to finish CAD, I'll re-

member the worksessions we 'got nothing done.' Because instead of machining, we were talking with one another. About the future of our team, brainstorming ideas to reach our potential, not the robot's, but the team's, the people on our team's. I see 'nerds' turn into dreamers, who care about more than just winning—about getting freshmen involved, breaking gender barriers, inspiring little kids. Taking notes on the whiteboard, talking, for hours. It's these work sessions that made me fall in love with robotics.



Machining Experience, Continued...

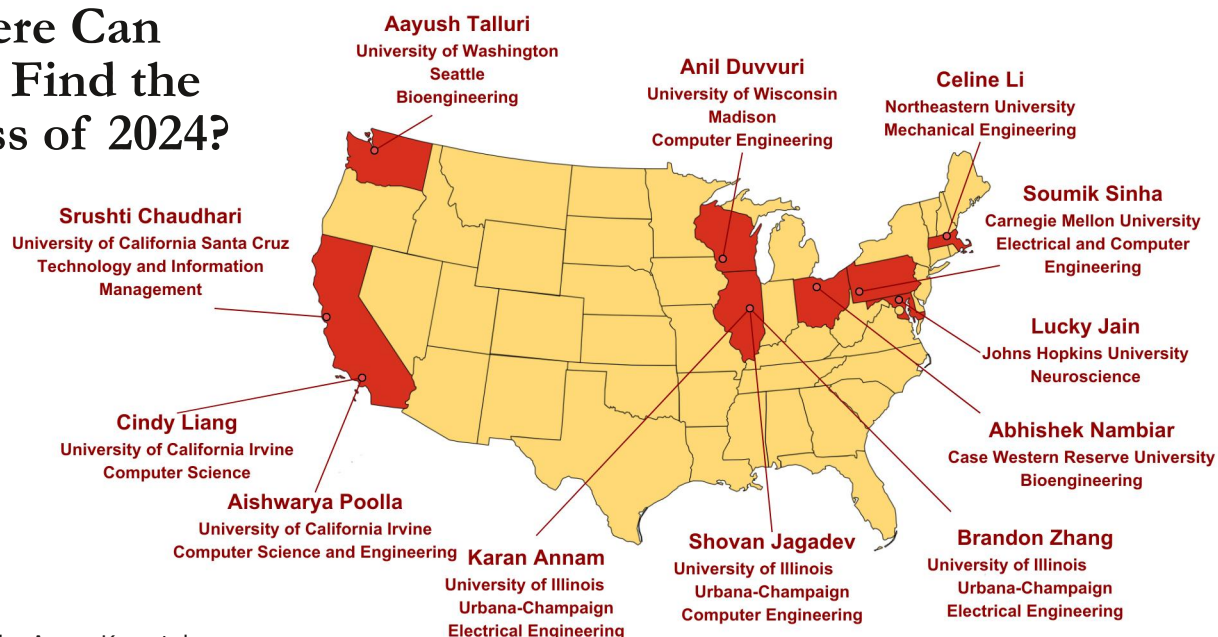
especially as a freshman. When I wasn't working in front of the CNC machines, I was helping out with the non-technical side of the team, such as by helping plan our teams' trip to Nampa for the Idaho Regional competition and Monkey Buddies, one of our numerous workgroups.

Over this past season, I slowly gained more confidence in my abilities and became more com-

fortable being on the team. Not only has it been a valuable experience where I could work in a tightly knit team community, but also something that has inspired me to learn more about engineering. Looking back, I can see that this team has not only provided me with a lot of new knowledge, but also allowed me to meet new friends and grow as a person.



Where Can You Find the Class of 2024?



Graphic by Arnav Kamatala